

BANA 4373: Advanced Business Analytics for Economics and Business

Lecture 3: Data Cleaning and Reproducibility in Practice

Dr. Fidel González

Sam Houston State University
Spring 2026

Agenda (and where we are in the weekly schedule)

This lecture corresponds to **Week 3** in the syllabus: **Data Preparation I (Part B): Cleaning and Reproducibility**.

Today we will:

- Review why cleaning is not “just formatting,” but a set of analytical decisions.
- Practice a workflow for quality checks (units, ranges, missing values, duplicates).
- Learn how to think about missing data (why it happens and what to do about it).
- Introduce identifiers and keys (why merges fail and how economists prevent it).
- Make your notebook reproducible: organization, documentation, and rerun-ability.

Big picture: We are building the habits you will need for Homework 2 and for every applied project in this class.

Why data preparation dominates real analytics

In real economics and business analytics, the main challenges are rarely “advanced math.” They are usually:

- **Dirty data:** missing values, wrong types, inconsistent units, typos in IDs, duplicate rows.
- **Unclear definitions:** variable meaning changes over time; metadata is ignored; measurement differs across sources.
- **Workflow problems:** manual edits, no documentation, no way to re-run the analysis.

Key idea: Models amplify errors; they do not fix them. If you feed a model the wrong data, you get precise-looking answers that are still wrong.

Today's mindset: “Before I compute anything, do I trust the data I am computing with?”

Data cleaning is an analytical decision

Cleaning is not neutral. Every choice embeds assumptions:

- **Which observations count?** (Do we drop counties with suppressed values? Do we keep them with flags?)
- **How do we treat missing values?** (Drop, impute, carry forward, or model missingness?)
- **Which definition applies?** (What year, what survey, what unit, what population?)

In economics and policy, these choices can change conclusions, for example:

- A trend can disappear after correcting an inflation adjustment.
- A “big difference” can be driven by a handful of outliers or by inconsistent measurement.
- A regression can flip signs if an ID mismatch merges the wrong units together.

Your job: Make decisions and **document them** so others can evaluate (and trust) your work.

Quick recap from Lecture 2 (what we already did)

Last time we introduced the idea of pulling data from an API (U.S. Census ACS):

- We formed an API request using a URL with variables and geography.
- We received JSON (a table in disguise) and loaded it into a pandas DataFrame.
- We learned an important practical detail: **API values often arrive as strings and must be converted to numeric types.**

Today's shift: Even if the data comes from a reputable API, we still need to verify:

- Are there missing or suppressed values?
- Do the numbers make sense in range and units?
- Are identifiers consistent and usable as keys?

A practical data quality checklist (use this every time)

Before analysis, run a checklist. In practice, you can do this in 3–5 minutes:

- ① **Structure:** What is a row? What is a column? Is this cross-section, time series, or panel?
- ② **Types:** Are numeric columns numeric? Are dates parsed as dates? Are IDs stored as strings?
- ③ **Missingness:** Which variables have missing values? How many? Is missingness clustered?
- ④ **Ranges and units:** Do min/max values make sense? Are units consistent (percent vs level)?
- ⑤ **Duplicates:** Are there duplicate rows? Duplicate IDs? Duplicate ID-time combinations?
- ⑥ **Keys:** What uniquely identifies a unit (e.g., state+county)? Does it actually behave uniquely?

Rule: If you cannot describe what a row represents and how it is identified, you are not ready to analyze.

Missing data: why it happens in economic datasets

Missing values are not random “mistakes.” They often reflect how the data is produced:

- **Survey nonresponse:** people or firms do not answer certain questions.
- **Confidentiality suppression:** agencies hide values for small samples to protect privacy.
- **Reporting thresholds:** data is not reported unless it meets minimum size or quality rules.
- **Measurement problems:** inconsistent reporting, classification changes, or coding errors.

Key question: Before choosing a method, ask: *Why is the value missing?*

In practice, the same numeric code can mean different things:

- Truly missing because “not collected”
- Missing because “not applicable”
- Missing because “suppressed for confidentiality”

Your solution should match the reason.

Handling missing data: a decision framework

There is no single “correct” method. Your choice depends on the context and the goal.

Common strategies:

- **Complete-case (drop missing):** simplest, but can bias results if missingness is systematic.
- **Simple imputation (mean/median):** easy, but can distort distributions and relationships.
- **Carry forward/backward (time series):** can be reasonable for slowly changing variables.
- **Missingness indicators:** keep the row and add a flag “missing = 1” so missingness becomes information.

What we care about in this class:

- You can explain your choice.
- You can show how sensitive results are (e.g., compare “drop” vs “flag”).
- You document the decision in your notebook.

Identifiers and keys: why economists obsess over them

A **key** is a variable (or set of variables) that uniquely identifies an observation.

Examples:

- **State-level cross-section:** state (FIPS code)
- **County-level cross-section:** state + county (FIPS codes)
- **County-by-year panel:** state + county + year

Why keys matter:

- Merges/join operations require reliable keys.
- Using names (e.g., “Washington County”) is dangerous because names repeat.
- IDs should often be stored as **strings** to preserve leading zeros (common in FIPS codes).

A common source of error: “My merge worked, but it merged the wrong units.”

This is one of the most expensive mistakes in applied work.

Even API data needs cleaning

It is tempting to think: “If it comes from a government API, it must be analysis-ready.” Not quite.

Common issues even in high-quality sources:

- **Types:** numeric values arrive as strings.
- **Suppression/missingness:** some counties or groups have suppressed estimates.
- **Metadata:** variable definitions and survey design matter.
- **Geography changes:** boundaries or coding conventions can change over time.

Practical takeaway: APIs reduce manual download errors, but they do not replace validation.

Data types and why they matter

When data enters Python (especially from APIs), variables may not have the type you expect.

Common data types in applied work:

- **String (text):** names, IDs, codes — but numbers often arrive this way from APIs.
- **Numeric (int, float):** values you can add, average, regress, or plot.
- **Categorical (conceptually):** labels like state, county, industry.
- **Datetime:** dates, months, quarters, years.

Why this matters:

- Sorting text numbers gives wrong order (“100” comes before “20”).
- Arithmetic fails or produces nonsense if values are stored as strings.
- Merges can silently fail if keys have different types.

Rule: Always check and explicitly set data types before analysis.

Data types and memory: a preview

Different data types use very different amounts of memory.

- **Strings** are flexible but memory-intensive.
- **Numeric types** are compact and fast.
- **Categorical variables** can dramatically reduce memory when values repeat.

For small datasets, this mainly affects **correctness**. For large datasets, it affects:

- speed
- memory usage
- whether your analysis runs at all

We will return to this later in the semester when we work with larger datasets and efficiency tools.

Mini case (in-class): ACS income and population

We will revisit the dataset we pulled: **Texas counties** with:

- Median household income (estimate)
- Total population (estimate)
- County and state identifiers

Questions we will ask before analysis:

- 1 Are there counties with missing income values? If so, why might that happen?
- 2 Do income values fall in a believable range? (What counts as “too low” or “too high”?)
- 3 If we compute income per capita (a rough measure), do we see implausible results?
- 4 Are county identifiers unique? (Does each county appear exactly once?)

Goal: Practice the workflow that turns “data pulled” into “data trusted.”

Reproducibility: from concept to habits

A result is reproducible if another person can start from raw inputs and get the same outputs using your code.

Reproducibility requires habits:

- **No manual spreadsheet edits:** every transformation should be in code.
- **Clear structure:** raw vs cleaned data; code vs output.
- **Documentation:** explain assumptions, units, and decisions in markdown.
- **Rerun-ability:** restart the kernel and “Run All” should work.

Why it matters:

- Credibility in academic work and policy analysis.
- Easier collaboration.
- Faster debugging.
- Your future self will thank you.

A simple project structure (recommended)

Use a consistent structure. This is not “extra”; it prevents errors.

```
project/  
  data_raw/          (never edit raw inputs)  
  data_clean/        (analysis-ready datasets created by code)  
  notebooks/         (Jupyter notebooks as reports)  
  src/               (optional: helper .py scripts)  
  output/            (figures, tables, exported results)  
  README.md          (how to run the project)
```

Rule: If you cannot explain your folder structure, you are probably doing manual steps that you will forget later.

Documentation: what to write in your notebook

Good notebooks are not just code. They are communication.

At minimum, each notebook should include:

- **Purpose:** What question is this notebook answering?
- **Data source:** Where did the data come from? (API URL, dataset name, year)
- **Variable notes:** What do the main variables mean? Units?
- **Cleaning decisions:** What did you drop/change and why?
- **Outputs:** What tables/figures did you produce and what do they mean?

Standard you should aim for: A classmate should be able to read your notebook and understand your workflow without asking you questions.

In-class workflow (what we will do in the notebook)

In the Jupyter notebook (based on our Lecture 2 work), we will:

- ① Pull county-level ACS data (or load the already-pulled DataFrame).
- ② Convert types (strings → numeric) and verify no silent errors.
- ③ Run data quality checks:
 - missing values summary
 - descriptive stats (min/max/median)
 - duplicates check for keys
- ④ Handle missingness in a transparent way:
 - compare “drop” vs “flag missing” approaches
- ⑤ Create at least one derived variable:
 - income per capita (rough) or log income
- ⑥ Save outputs in a reproducible way (no copy/paste).

Deliverable mindset: You are building a clean dataset that is ready for merging and analysis next week.

Common mistakes to avoid (and why they matter)

These are the mistakes that cost the most time and lead to incorrect conclusions:

- **Not checking types:** numbers stored as strings break calculations or silently coerce.
- **Treating missing as zero:** zero is a real value; missing is “unknown.”
- **Using names as identifiers:** names repeat and spellings vary.
- **Ignoring units:** percent vs proportion; nominal vs real; monthly vs annual.
- **Manual edits:** anything you do by hand you cannot reliably reproduce.

Professional habit: Always write code and always explain what the code is doing in plain language.

Looking ahead (how this connects to the next weeks)

Next we will build directly on today:

- **Week 4:** Data Preparation II — Merging/joining datasets using reliable keys.
- **Week 5:** Building a panel dataset (county-by-month or county-by-year).
- **Week 6:** Visualization fundamentals: telling a clear economic story with figures.

Why today is essential: If your cleaned dataset is not trustworthy, every step after this becomes fragile (and sometimes meaningless).

Key takeaways

- Cleaning is an analytical decision, not a cosmetic step.
- Missing data is rarely random; ask why it is missing before deciding what to do.
- Identifiers and keys are the foundation of merges; incorrect keys can quietly destroy an analysis.
- Reproducibility is credibility: structure, documentation, and rerun-ability matter.
- The goal is not “perfect data.” The goal is transparent and defensible choices.

Next class: we begin merges and reshaping, using the clean dataset you create today.

End

Questions?